Problem 1

Consider an m-by-n chessboard in which certain squares are forbidden. What is the largest number of nonattacking rooks that can be placed on the board?

Problem 2

Consider again an m-by-n chessboard where certain squares are forbidden. What is the largest number of dominoes that can be placed on the board so that each domino covers two allowed squares and no two dominoes overlap (cover the same square)?

Problem 3

A company has n jobs available, with each job demanding certain qualifications. There are m people who apply for the n jobs. What is the largest number of jobs that can be filled from the available m applicants if a job can be filled only by a person who meets its qualifications?



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General problem formulation I

- Let

$$X = \{x_1, x_2, \ldots, x_m\}$$

and

$$Y = \{y_1, y_2, \ldots, y_n\}$$

be two finite sets with m elements and n elements, respectively. We assume that the sets X and Y have no elements in common, that is,

$$X \cap Y = \emptyset.$$

Let Δ be a collection of pairs

$$e = \{x, y\},$$



General problem formulation II

where x is an element of X and y is an element of Y. The triple

$$G = (X, \Delta, Y)$$

is called a *bipartite graph*. The elements of $X \cup Y$ are called the *vertices* of G, and X, Y is called a bipartition (partition into two parts) of the vertices of G. The pairs $e = \{x, y\}$ in Δ are called the *edges* of G. We say that the edge e *joins* the vertices x and y, and that the vertices x and y *meet* the edge e.

- A bipartite graph is prescribed by
  1. a set of vertices;
  2. a partition of that set of vertices into two parts; and
  3. a set of edges joining a vertex in on part to a vertex in the other part.



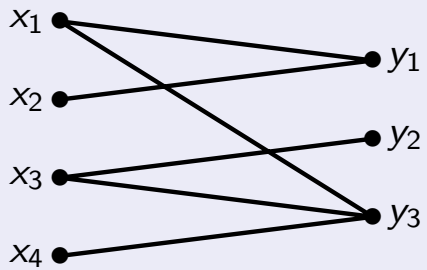
General problem formulation III

Example

Let $X = \{x_1, x_2, x_3, x_4\}$ and $Y = \{y_1, y_2, y_3\}$, and let

$$\Delta = \{\{x_1, y_1\}, \{x_1, y_3\}, \{x_2, y_1\}, \{x_3, y_2\}, \{x_3, y_3\}, \{x_4, y_3\}\}.$$

The bipartite graph $G = (X, \Delta, Y)$ is shown as



- As a result, we now speak of the vertices of X as the *left vertices* and the vertices of Y as the *right vertices*.



General problem formulation IV

- Let $G = (X, \Delta, Y)$ be a bipartite graph. A *matching* of G is defined to be a subset M of the set Δ of edges, with the property that no two of the edges of M have a common vertex. Thus, if M is a matching, then each left vertex meets at most one edge of M, and similarly each right vertex meets at most one edge of M.
- In the previous example, the three edges

$$\{x_1, y_3\}, \{x_2, y_1\}, \{x_3, y_2\}$$

form a matching of size three.

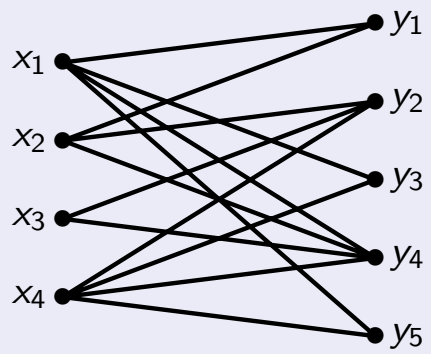


General problem formulation V

Example

Consider the 4-by-5 board with forbidden positions,

	y_1	y_2	y_3	y_4	y_5
x_1		×			
x_2			×		×
x_3	×		×		×
x_4	×				



We associate a bipartite graph G with it, in which x_i is the left vertex corresponding to row i and y_j is the right vertex corresponding to column j. We join x_i and y_j by an edge if and only if the square at the intersection of row i and column j is allowed.



General problem formulation VI

- Consider the matching

$$M = \{\{x_1, y_1\}, \{x_2, y_4\}, \{x_4, y_2\}\}.$$

  If we put rooks on the three squares corresponding to the edges of M, we have three nonattacking rooks on the board. Conversely, a collection of nonattacking rooks on the board gives a matching of the bipartite graph.
- We can associate a bipartite graph $G = (X, \Delta, Y)$, called a *rook-bipartite graph*, with any m-by-n board B with forbidden positions.
- This graph has vertices $X = \{x_1, x_2, \ldots, x_m\}$ corresponding to the rows of the board, and vertices $Y = \{y_1, y_2, \ldots, y_n\}$ corresponding to the columns. The pair $\{x_i, y_j\}$ is an edge of Δ if and only if the square at the intersection of row i and column j is allowed.
- Nonattacking rooks on the board B correspond to matchings in the bipartite graph G.



General problem formulation VII

- Moreover, the largest number of nonattacking rooks that can be placed on the board B equals the largest number of edges in a matching of the bipartite graph G. Each bipartite graph is the rook-bipartite graph of some board with forbidden positions.
- If G is any bipartite graph, we now define

$$\rho(G) = \max\{|M| : M \text{ a matching}\}$$

to be the size of the largest matching M of G.

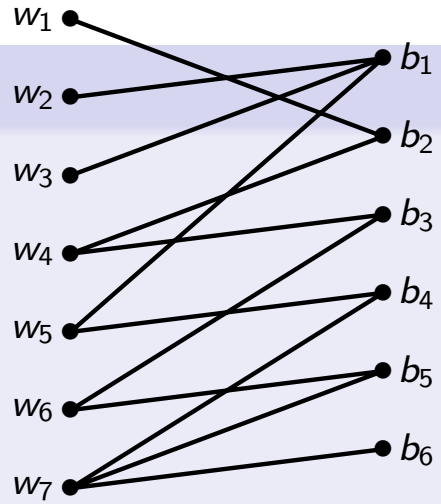


General problem formulation VIII

- Problem 1 is equivalent to determining $\rho(G)$ for the rook-bipartite graph corresponding to a board with forbidden positions.

Example

w_1	$\times$	w_2	b_1	w_3
b_2	w_4	$\times$	w_5	$\times$
$\times$	b_3	$\times$	b_4	$\times$
$\times$	w_6	b_5	w_7	b_6



- There is a one-to-one correspondence between sets of nonoverlapping dominoes on the board and matchings in the bipartite graph associated with the board.



General problem formulation IX

- Given an m-by-n board B with forbidden positions, we associate a *domino-bipartite* graph $G = (X, \Delta, Y)$ where

  $$X = \{w_1, w_2, \ldots, w_p\}$$

  is the set of white squares,

  $$Y = \{b_1, b_2, \ldots, b_q\}$$

  is the set of black squares, and there is an edge $\{w_i, b_j\}$ in Δ if and only if one domino can simultaneously cover both w_i and b_j.
- Nonoverlapping dominoes on the board B correspond to matching edges in the bipartite graph G. The largest number of nonoverlapping dominoes that can be placed on the board equals the largest number $\rho(G)$ of edges in a matching of G.



General problem formulation X

- Not every bipartite graph is the domino-bipartite graph of a board with forbidden positions.

Example

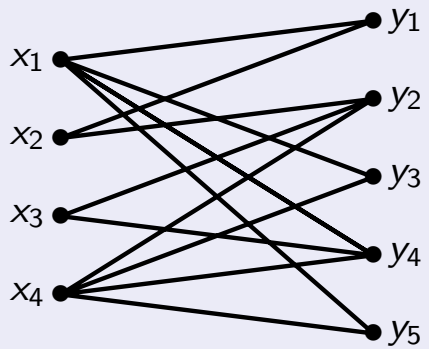
Four people x_1, x_2, x_3, x_4 apply for five jobs y_1, y_2, y_3, y_4, y_5. Suppose that

x_1 is qualified for y_1, y_3, y_4, y_5;

x_2 is qualified for y_1, y_2, y_4;

x_3 is qualified for y_2, y_4 and

x_4 is qualified for y_2, y_3, y_4, y_5.



- In sum, all three of the introductory problems are concerned with the problems of determining the largest number $\rho(G)$ of edges of a matching in a bipartite graph (X, Δ, Y).



Matchings I

- We consider a bipartite graph

$$G = (X, \Delta, Y)$$

where

$$X = \{x_1, x_2, \ldots, x_m\}, \qquad Y = \{y_1, y_2, \ldots, y_n\}.$$

Our goal is not only to determine $\rho(G)$, but also to determine a matching M^* with

$$|M^*| = \rho(G). \tag{9.1}$$

- By the pigeonhole principle, we have the simple inequality

$$\rho(G) \leq \min\{m, n\}. \tag{9.2}$$

Each matching M satisfies $|M| \leq \rho(G)$.

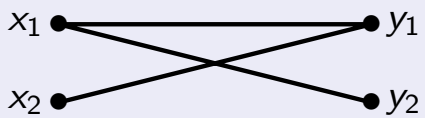


Matchings II

- A matching M^* satisfying (9.1)–that is, a matching M with the largest possible numbers of edges among all matchings in G–is called a *max-matching*.

Example

Consider the bipartite graph G. The edges $\{x_1, y_2\}$ and $\{x_2, y_1\}$ form a matching of size two and hence, we have $\rho(G) = 2$. Notice that the edge $\{x_1, y_1\}$ determines a matching M with one edge. There is no way to add another edge to this matching M in order to obtain a matching with two edges.



Matchings III

- Let u and v be two vertices in the bipartite graph $G = (X, \Delta, Y)$. A *path* joining u and v is a sequence of distinct vertices (except that u may equal v)

$$\gamma : u = u_0, u_1, u_2, \cdots, u_{p-1}, u_p = v \tag{9.3}$$

such that any two consecutive vertices are joined by an edge. Thus, in order for (9.3) to be a path,

$$\{u_0, u_1\}, \{u_1, u_2\}, \ldots, \{u_{p-1}, u_p\} \tag{9.4}$$

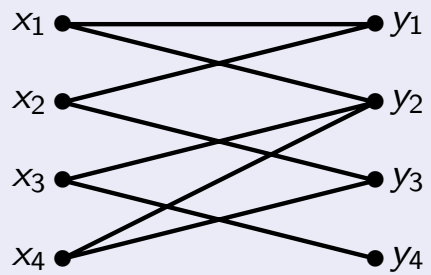
must all be edges in Δ. The edges in (9.4) are called the *edges of the path* γ. The *length* of the path γ is the number p of its edges. The vertices u and v are called the *end-vertices* of the path γ. If $u = v$ in the path (9.4), then the path is called a *cycle*. — A cycle in a bipartite graph necessarily has even length.



Matchings IV

Example

In the bipartite graph G pictured in Figure



$$x_1, y_2, x_3, y_4$$

is a path of length 3 joining x_1 and y_4;

$$y_1, x_2, y_3, x_4$$

is a path joining y_1 and x_4;

$$x_3, y_2, x_4, y_3, x_2$$

is a path joining x_3 and x_2. Also

$$x_1, y_1, x_2, y_3, x_4, y_2, x_1$$

is a cycle of length 6.

Matchings V

- Now let M be a matching in the bipartite graph $G = (X, \Delta, Y)$. Let $\overline{M}$ be the complement of M. Let u and v be vertices, where one of u and v is a left vertex and the other is a right vertex. A path γ joining u and v is an *alternating path with respect to the matching* M (for brevity, an *M-alternating path*) provided that the following properties hold,
  1. The first, third, fifth, ... edges of γ do not belong to the matching M (thus they belong to $\overline{M}$).
  2. The second, fourth, sixth, ... edges of γ belong to the matching M.
  3. Neither u nor v meets an edge of the matching M.



Matchings VI

- Notice that the length of an M-alternating path γ is an odd number $2k+1$ with $k \geq 0$, and that $k+1$ of the edges of γ are edges of $\overline{M}$ while k of the edges of γ are edges of M. We introduce further notation as follows,

  M_γ *denotes those edges of* γ *that belong to* M*;*

  and

  $\overline{M}_\gamma$ *denotes those edges of* γ *that do not belong to* M*.*

  It follows that

$$|\overline{M}_\gamma| = |M_\gamma| + 1.$$

Example



Matchings VII

$$M = \{\{x_1, y_1\}, \{x_2, y_3\}, \{x_3, y_4\}\}$$

is a matching of three edges.

$$\gamma : u = x_4, y_3, x_2, y_1, x_1, y_2 = v$$

is an M-alternating path. We have

$$M_\gamma = \{\{x_2, y_3\}, \{x_1, y_1\}\}, \quad \overline{M}_\gamma = \{\{x_4, y_3\}, \{x_2, y_1\}, \{x_1, y_2\}\}.$$

and we can obtain a matching of four edges

$$\begin{aligned} M' =& (M - M_\gamma) \cup \overline{M}_\gamma \\ =& \{\{x_3, y_4\}, \{x_4, y_3\}, \{x_2, y_1\}, \{x_1, y_2\}\}. \end{aligned}$$



Matchings VIII

Theorem

Let M be a matching in the bipartite graph $G = (X, \Delta, Y)$. Then M is a max-matching if and only if there does not exist an M-alternating path.

- As observed above, if M is a max-matching, then there does not exist an M-alternating path.
- To establish the converse, we now assume that M is not a max-matching and prove that there exists an M-alternating path. Let M' be a matching satisfying

$$|M'| > |M|.$$



Matchings IX

- We consider the bipartite graph

$$G^* = (X, \Delta^*, Y)$$

where

$$\Delta^* = (M - M') \cup (M' - M).$$

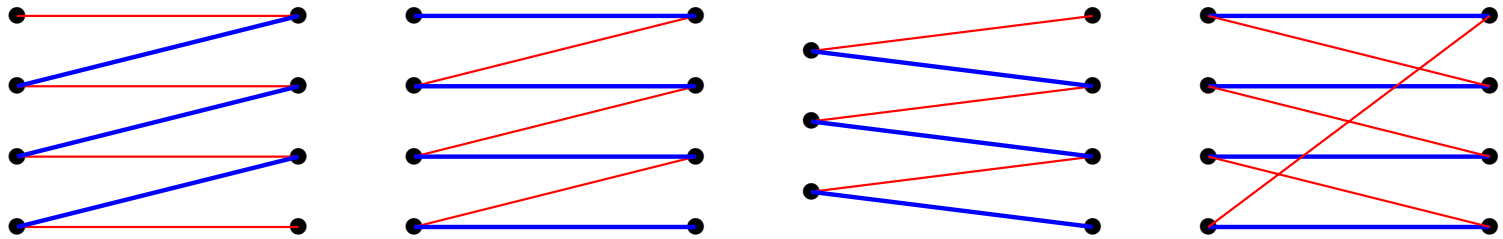
Thus, to get G^* we remove from G all edges that belong neither to M nor to M' and also those edges that belong to both M and M'. We have

$$|M' - M| > |M - M'|. \tag{9.5}$$

- The bipartite graph G^* has the property that each of its vertices meets, at most, two edges; each vertex meets, at most, one edge of $M - M'$ since M is a matching, and at most, one edge of $M' - M$ since M' is a matching.
- It implies that the set of edges of G^* can be partitioned into paths and cycles. In each of paths and cycles of this partition, the edges alternate between $M - M'$ and $M' - M$.



Matchings X



Type 1 Type 2 Type 3 Type 4

- By (9.5), there must be a path of Type 1. But a path of Type 1 is by definition an M-alternating path. □
- Let $G = (X, \Delta, Y)$ be a bipartite graph. A subset S of the set $X \cup Y$ of vertices of G is called a *cover*, provided that each edge of G has at least one of its two vertices in S,

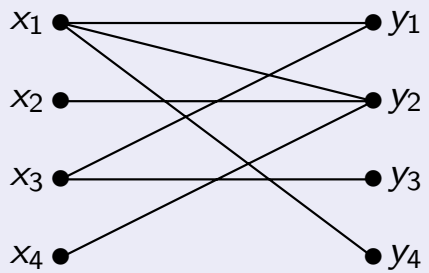
$$\{x, y\} \cap S \neq \emptyset \text{ for all } \{x, y\} \text{ in } \Delta.$$

Matchings XI

- The set X of left vertices of G is a cover. The set Y of right vertices is also a cover. However, our interest lies in small covers.

Example

In addition to the covers $\{x_1, x_2, x_3, x_4\}$ and $\{y_1, y_2, y_3, y_4\}$, we have the cover $S = \{x_1, x_3, y_2\}$ with only three vertices. The fact that S is a cover means that there is no edge whose left vertex is one of $\{x_2, x_4\}$ and whose right vertex is one of $\{y_1, y_3, y_4\}$.



Matchings XII

- We define the *cover number* of G to be

$$c(G) = \min\{|S| : S \text{ a cover of } G\},$$

the smallest number of vertices in a cover of G. Every cover S satisfies

$$|S| \geq c(G).$$

We call a cover S of G that satisfies

$$|S| = c(G)$$

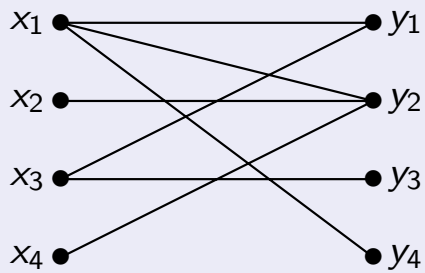
a *min-cover*.



Matchings XIII

Example

$S = \{x_1, x_3, y_2\}$ is a cover of G. It is straightforward to check that $c(G) = 3$ and S is a min-cover. If we do not include x_1 in a cover, then each of the three vertices y_1, y_2, and y_4 would have to be in the cover. If we do not include y_2 in the cover, we would have to include each of the three vertices x_1, x_2, and x_4 in the cover. Since $\{x_1, y_2\}$ is not a cover, each cover contains at least three vertices.



Lemma

If G is a bipartite graph, then

$$\rho(G) \leq c(G), \tag{9.6}$$

that is, the largest number of edges in a matching of G does not exceed the smallest number of vertices in a cover of G.

Matchings XIV

Proof

Let G be the bipartite graph (X, Δ, Y) and let S^* be a cover satisfying $S^* = c(G)$. Let M be a matching. Each edge of M has at least one of its vertices in S. Suppose that $|M| > |S^*|$. By the pigeonhole principle, two different edges in M contain the same vertex of S^*. But this contradicts the fact that M is a matching. Hence $|M| \leq |S^*| = c(G)$.



Matchings XV

Example

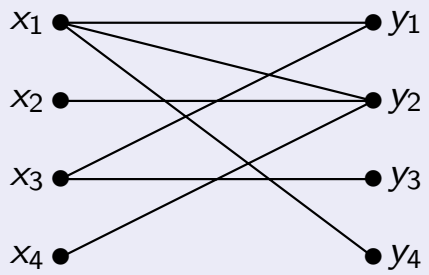
We see that

$$M = \{\{x_1, y_1\}, \{x_2, y_2\}, \{x_3, y_3\}\}$$

is a matching of three edges. As already observed, $S = \{x_1, x_3, y_2\}$ is a cover of three vertices. Hence,

$$3 = |M| \leq \rho(G) \leq c(G) \leq |S| = 3.$$

We have equality throughout, and hence M is a max-matching, S is a min-cover, and $\rho(G) = c(G) = 3$.



Matching Algorithm

Let $G = (X, \Delta, Y)$ be a bipartite graph where $X = \{x_1, x_2, \ldots, x_m\}$ and $Y = \{y_1, y_2, \ldots, y_n\}$. Let M be any matching in G.

Matchings XVI

(0) Begin by labeling with (*) all vertices in X that do not meet any edge in M and call all vertices *unscanned*. Go to (1).

(1) If in the previous step, no new label has been given to a vertex of X, then stop. Otherwise go to (2).

(2) While there exists a labeled, but unscanned, vertex of X, select such a vertex, say, x_i, and label with (x_i) all vertices in Y joined to x_i by an edge not belonging to M not previously labeled. The vertex x_i is now scanned. If there are no labeled but unscanned vertices go to (3).

(3) If, in step (2), no new label has been given to a vertex of Y, then stop. Otherwise go to (4).

(4) While there exists a labeled, but unscanned vertex, of Y, select such a vertex, say, y_j, and labeled with (y_j) any vertex of X joined to y_j by an edge belonging to M not previously labeled. The vertex y_j is now scanned. If there are no labeled but unscanned vertices, go to (1).



Matchings XVII

- There are two possibilities to consider:

  Breakthrough: *There is a labeled vertex of Y that does not meet an edge of M.*

  Non-Breakthrough: *The algorithm has come to a halt and Breakthrough has not occurred; that is, each vertex of Y that is labeled also meets some edge of M*

Theorem

Assume Non-Breakthrough occurs in the Matching Algorithm. Let X^{un} consist of all the unlabeled vertices of X and Y^{lab} consist of all the labeled vertices of Y. Then both of the following hold,

1. $S = X^{un} \cup Y^{lab}$ is a min-cover of the bipartite graph G;
2. $|M| = |S|$ and M is a max-matching.

- Assume that there is an edge $e = \{x, y\}$ neither of its vertices belongs to S. Thus, x is in $X - X^{un}$ and y is in $Y - Y^{lab}$. Namely, x is labeled and y is unlabeled.



Matchings XVIII

- If e does not belong to M, in applying step (2) of the algorithm, y receives the label x, a contradiction.
- If e belongs to M, by step (0), the label of x is not $(*)$. But then it follows that x has label (y) by step (4). Vertex y can give label (y) to a vertex of X only if y is labeled. Since y is unlabeled, we have a contradiction again.
- We conclude that S is a cover.
- Let y be a vertex in Y^{lab} so that y is labeled. Since Breakthrough has not occurred, y meets an edge of M, and hence, exactly one edge, say, the edge $\{x, y\}$ of M.
- By step (4), x gets the label (y) and hence x is not in X^{un}. Thus, each vertex of Y^{lab} meets an edge of M whose other vertex belongs to $X - X^{un}$.
- Consider a vertex x' in X^{un}. By step (0), x' meets an edge of M, and hence exactly one edge, say, $\{x', y'\}$, of M. The vertex y' can not be in Y^{lab} since the unique edge of M meeting a vertex in Y^{lab} has its other vertex in $X - X^{un}$.



Matchings XIX

- Thus, for each vertex in $X^{un} \cup Y^{lab}$ there is a unique edge of M containing it and all these edges are distinct. Hence

$$|S| = |X^{un} \cup Y^{lab}| = |M|,$$

and we conclude that $|S| = |M|$. □

Corollary

[König's theorem] Let $G = (X, \Delta, Y)$ be a bipartite graph. Then

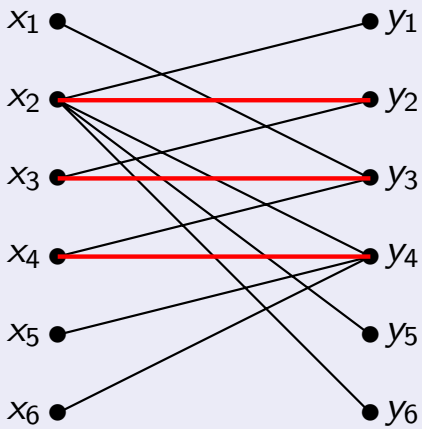
$$\rho(G) = c(G);$$

that is, the largest number of edges in a matching equals the smallest number of vertices in a cover.

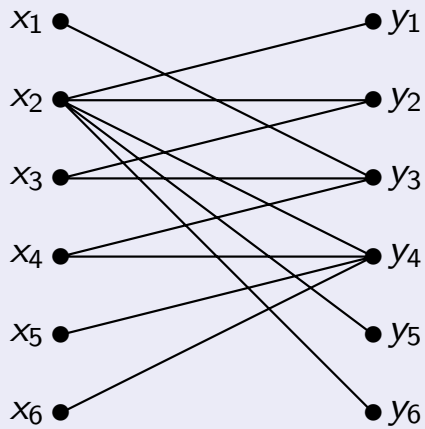


Matchings XX

Example



(a)

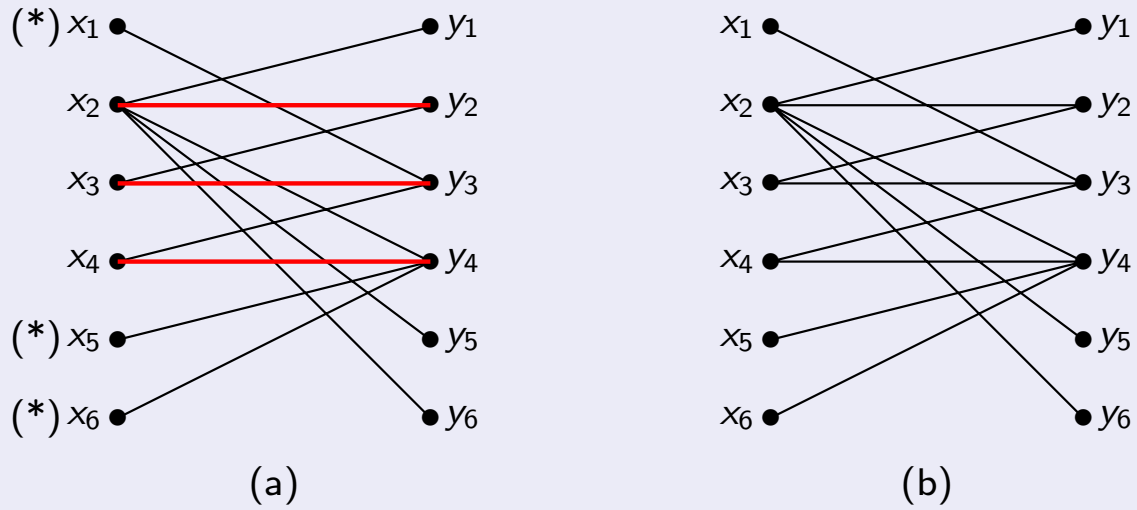


(b)

We begin with $M^1 = \{\{x_2, y_2\}, \{x_3, y_3\}, \{x_4, y_4\}\}$, as well as the labeling of the vertices, shown in (a).

Matchings XX

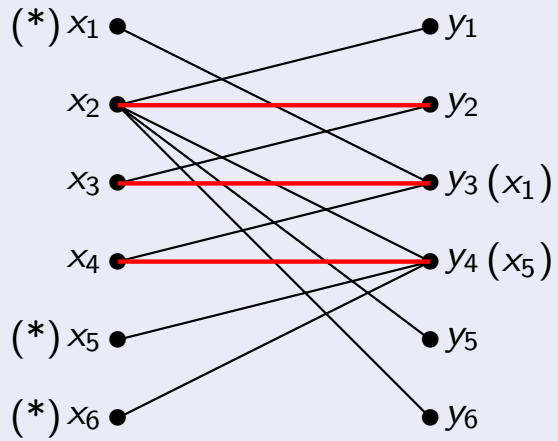
Example



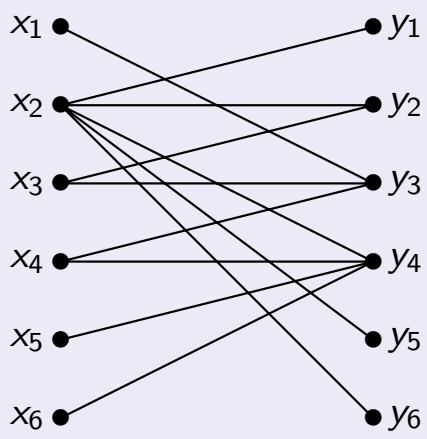
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Matchings XX

Example



(a)

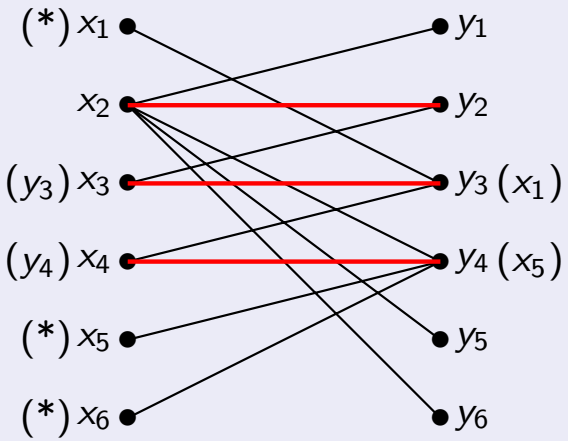


(b)

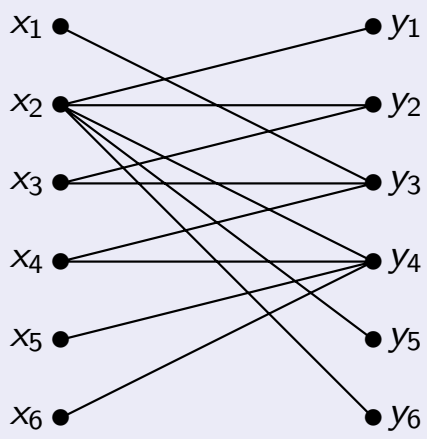
We begin with $M^1 = \{\{x_2, y_2\}, \{x_3, y_3\}, \{x_4, y_4\}\}$, as well as the labeling of the vertices, shown in (a).

Matchings XX

Example



(a)

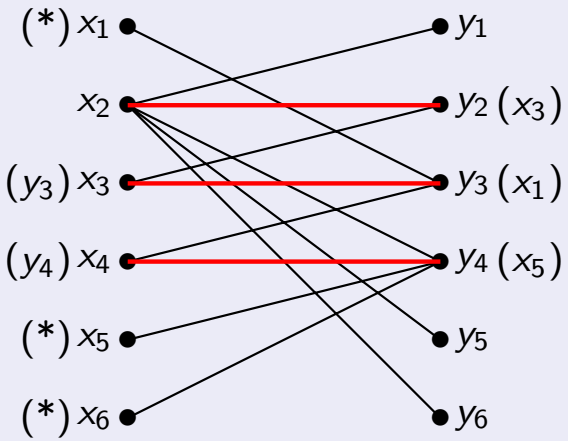


(b)

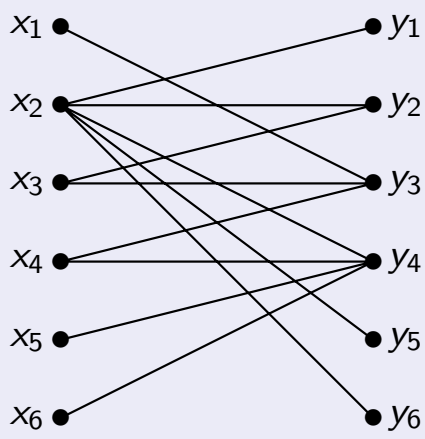
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Matchings XX

Example



(a)

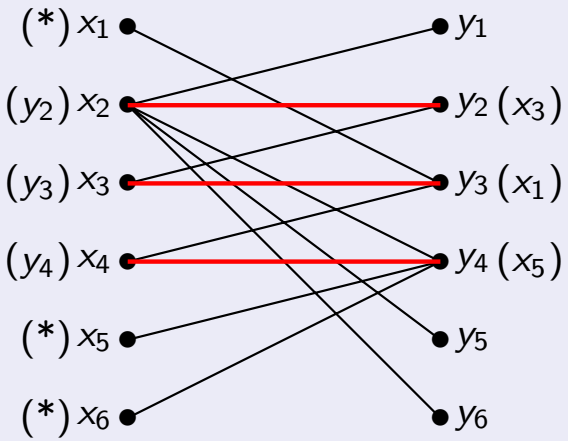


(b)

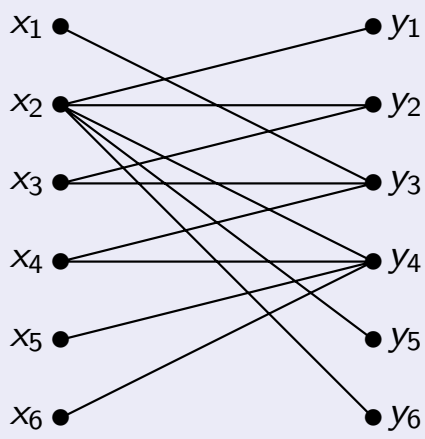
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Matchings XX

Example



(a)

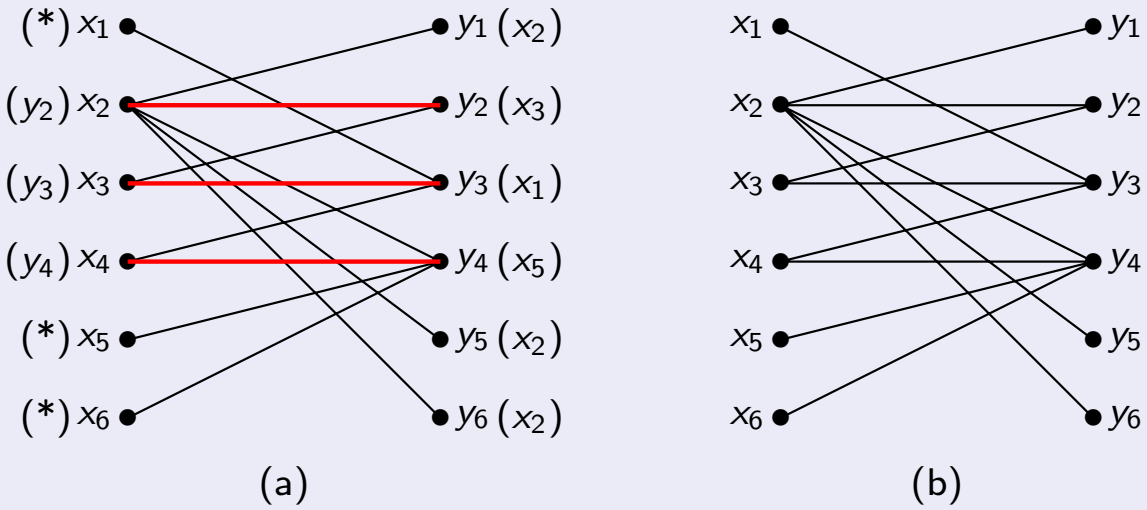


(b)

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Matchings XX

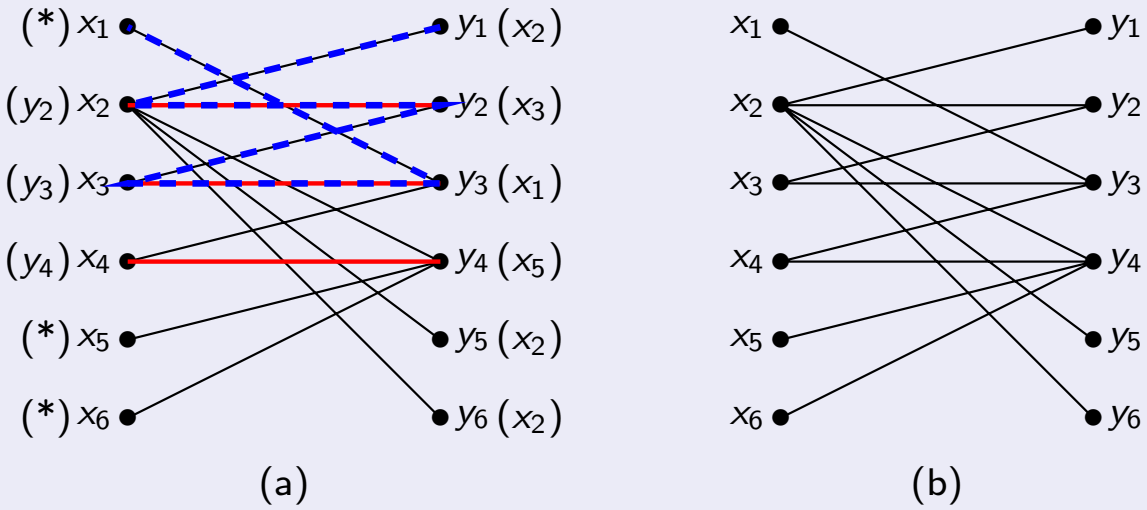
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Matchings XX

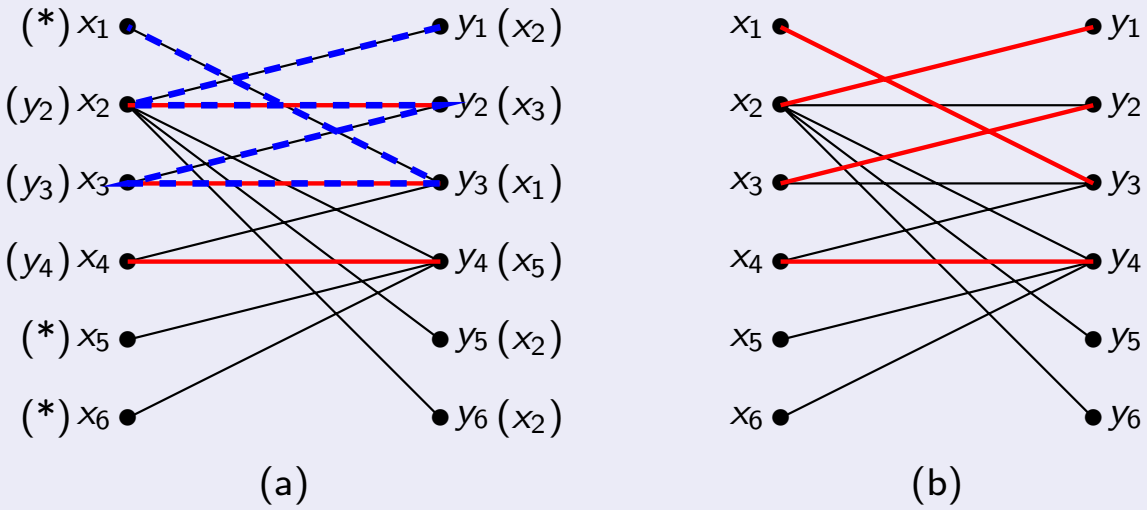
Example



We begin with $M^1 = \{\{x_2, y_2\}, \{x_3, y_3\}, \{x_4, y_4\}\}$, as well as the labeling of the vertices, shown in (a). We find the M^1-alternating path $\gamma : y_1, x_2, y_2, x_3, y_3, x_1$.

Matchings XX

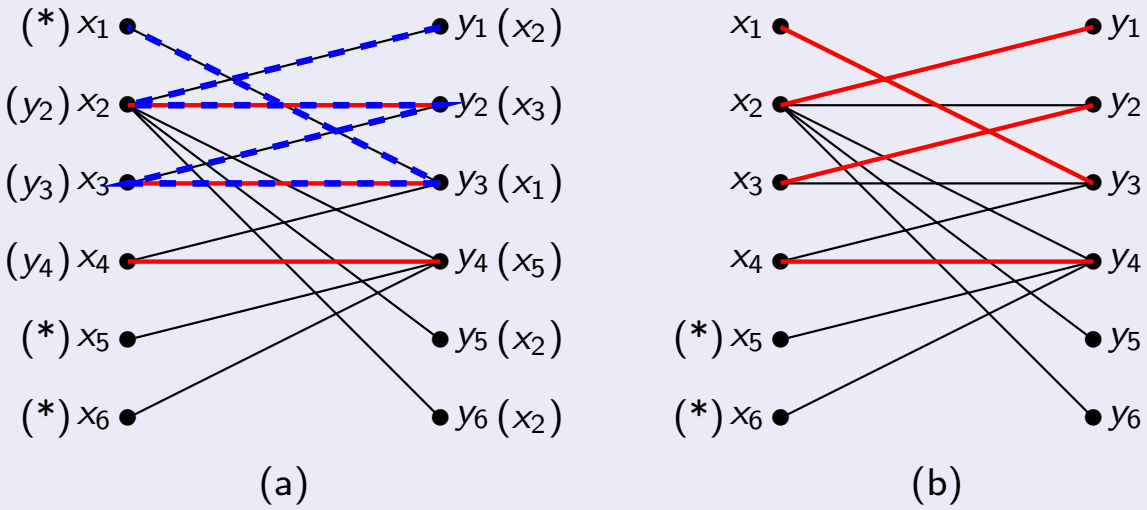
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Matchings XX

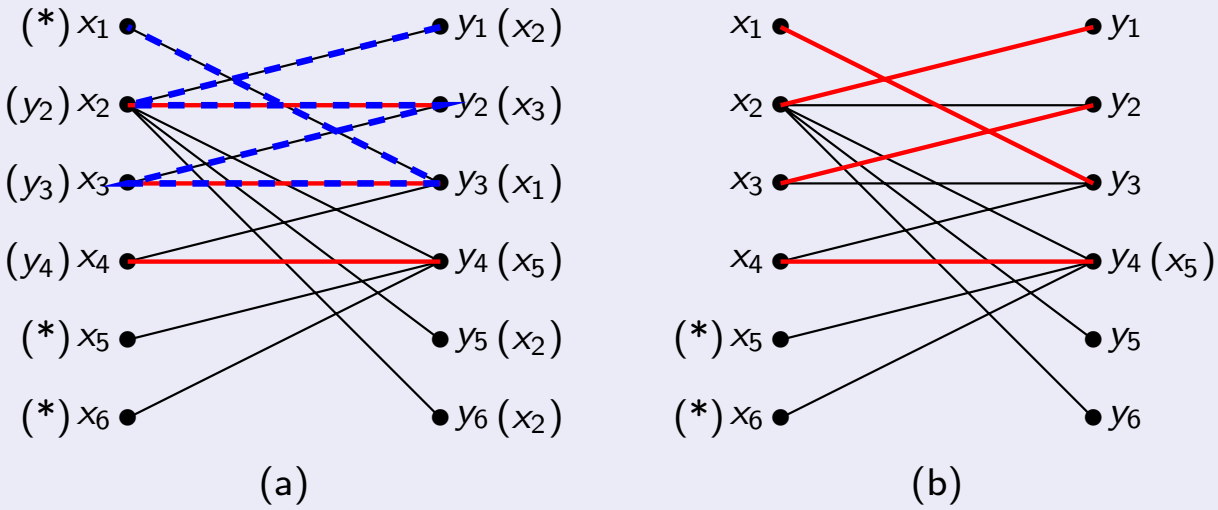
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Matchings XX

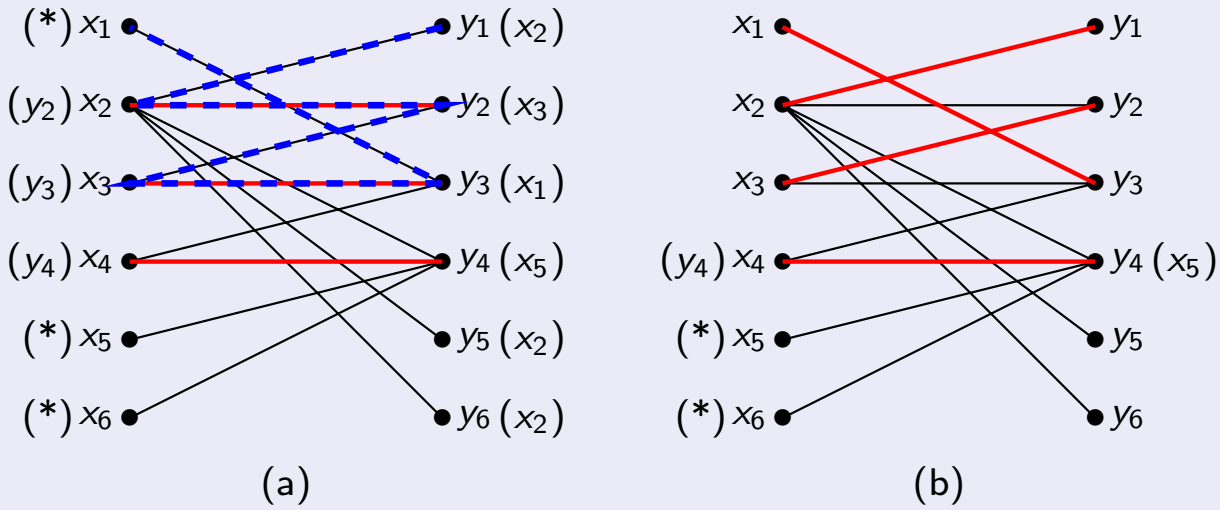
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Matchings XX

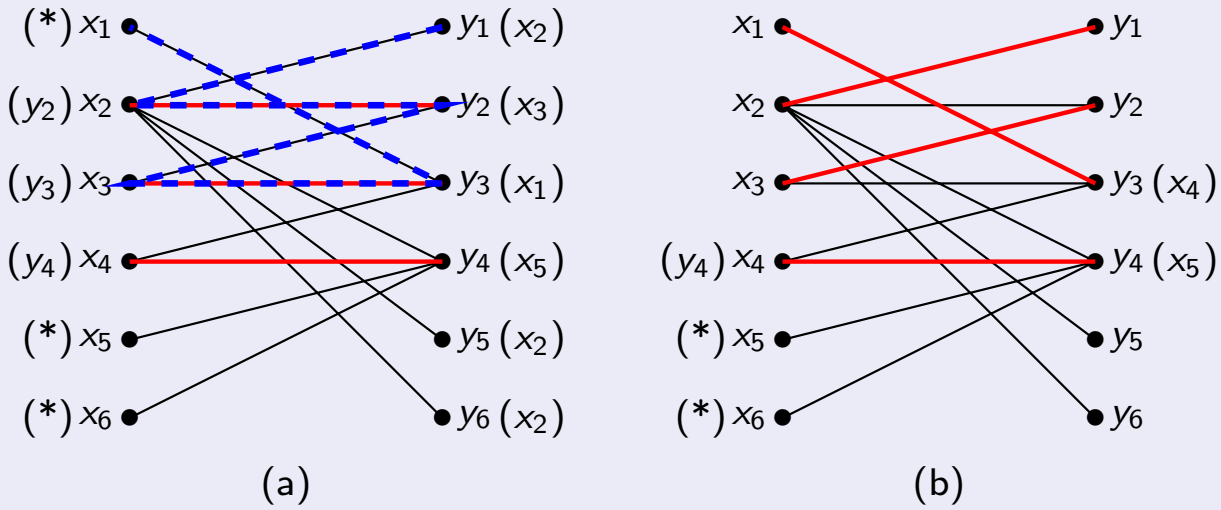
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Matchings XX

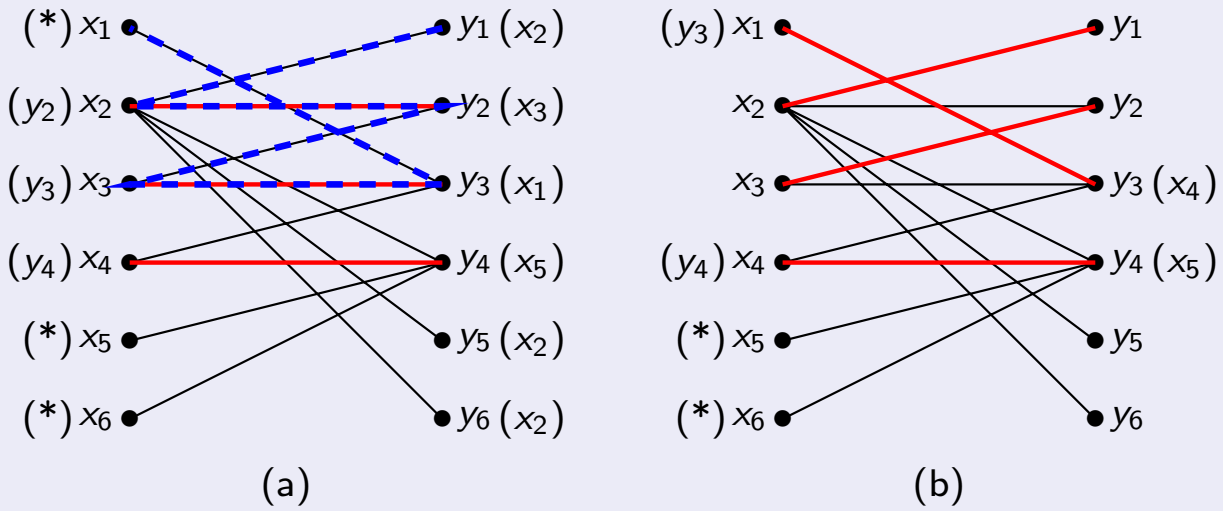
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Matchings XX

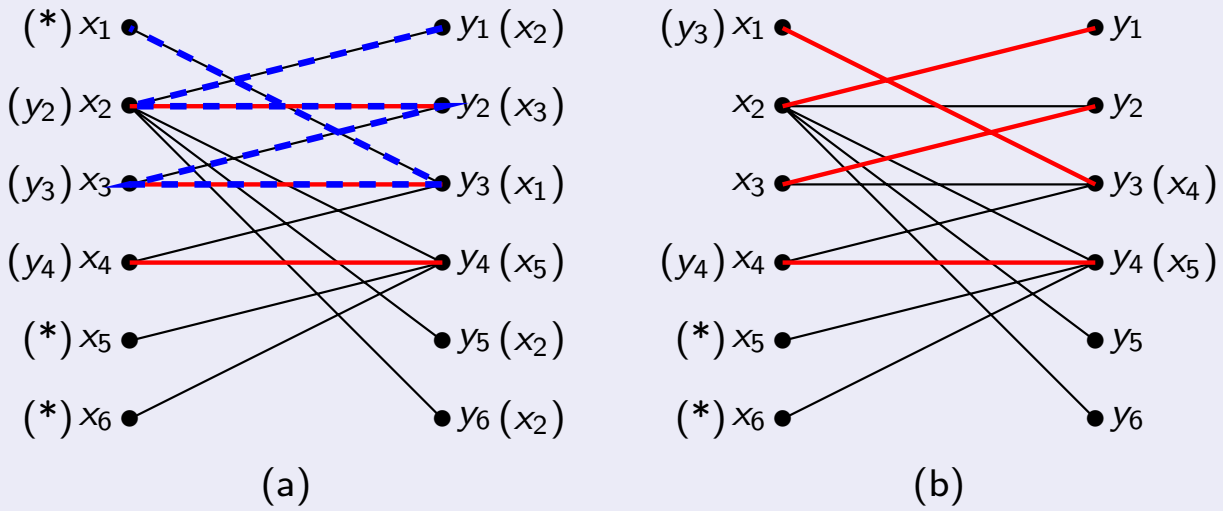
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Matchings XX

Example



We begin with $M^1 = \{\{x_2, y_2\}, \{x_3, y_3\}, \{x_4, y_4\}\}$, as well as the labeling of the vertices, shown in (a). We find the M^1-alternating path $\gamma : y_1, x_2, y_2, x_3, y_3, x_1$. Then $M^2 = \{\{x_4, y_4\}, \{y_1, x_2\}, \{y_2, x_3\}, \{y_3, x_1\}\}$ is a matching of four edges, as well as the labeling of the vertices, shown in (b). The breakthrough has not occurred in this case. The set $S = \{x_2, x_3, y_3, y_4\}$ of size 4, consisting of the unlabeled vertices of X and the labeled vertices of Y, is a min-cover.

Matchings XXI

- Let $G = (X, \Delta, Y)$ be a bipartite graph such that the set X of left vertices and the set Y of right vertices have the same size n. A matching in G can contain at most n edges. A matching M in G with n edges is called a *perfect matching*.
- It follows from the corollary that G has a perfect matching if and only if no set of fewer than n vertices covers all the edges of G.
- A bipartite graph $G = (X, \Delta, Y)$ is called *regular of degree* p, provided that each of its vertices meets exactly p edges. If G is regular of degree $p \geq 1$, then X and Y must have the same size n.

Theorem

A bipartite graph $G = (X, \Delta, Y)$ that is regular of degree $p \geq 1$ always has a perfect matching.



Matchings XXII

Proof

Let X and Y each contain n vertices. Let S be any cover of G. Every edge of G meets at least one vertex of S. Since G is regular of degree p each of vertex of S meets exactly p edges. Hence the total number of edges of G is at most $p|S|$. But the total number of edges of S is pn. Hence $p|S| \geq pn$ and thus $|S| \geq n$. Therefore every cover of G has at least n vertices and G has a perfect matching.

Example

At a party there are n boys and n girls. Suppose that there exists a positive integer p such that each boy has been previously introduced to p girls, and each girl has been previously introduced to p boys. Show that the boys and girls can be paired up so that the boy and girl of each pair are acquainted.

Let $G = (X, \Delta, Y)$ be constructed as follows. The set X of left vertices consists of the n boys, and the set Y of the right vertices consists of the n girls. We join a boy and a girl by an edge if and only if they are acquainted. The assumption implies that G is regular of positive degree p. G has a perfect matching, which describes the required pairing.



Systems of distinct representatives I

- Let Y be a finite set, and let $\mathcal{A} = \{A_1, A_2, \ldots, A_n\}$ be a family of n subsets of Y. A family $(e_1, e_2, \ldots, e_n)$ of elements of Y is called a *system of representatives of* $\mathcal{A}$, provided that e_1 is in A_1, e_2 is in A_2, ..., e_n is in A_n.
- If, in a system of representatives, the elements $e_1, e_2, \ldots, e_n$ are all different, then $(e_1, e_2, \ldots, e_n)$ is called a *system of distinct representatives*, abbreviated SDR.

Example

Let $\{A_1, A_2, A_3, A_4\}$ be the family of subsets of the set $Y = \{a, b, c, d, e\}$, defined by

$$A_1 = \{a, b, c\}, A_2 = \{b, d\}, A_3 = \{a, b, d\}, A_4 = \{b, d\}.$$

Then $\{a, b, b, d\}$ is a system of representatives, and $\{c, b, a, d\}$ is an SDR.



Systems of distinct representatives II

Example

Let the family $\mathcal{A} = \{A_1, A_2, A_3, A_4\}$ be defined by

$$A_1 = \{a, b\}, A_2 = \{a, b\}, A_3 = \{a, b\}, A_4 = \{a, b, c, d\}.$$

Then $\mathcal{A}$ does not have an SDR.

Lemma

In order for the family $\mathcal{A} = \{A_1, A_2, \ldots, A_n\}$ of sets to have an SDR, it is necessary that the following condition hold,

(MC): For each $k = 1, 2, \ldots, n$ and each choice of k distinct indices $i_1, i_2, \ldots, i_k$ from $\{1, 2, \ldots, n\}$,

$$|A_{i_1} \cup A_{i_2} \cup \cdots \cup A_{i_k}| \geq k. \tag{9.7}$$

Condition (MC) in the lemma is often called the *Marriage Condition*.



Systems of distinct representatives III

Example

The Marriage Problem – There are n men and m women, and all the men are eager to marry. If there were no restrictions on who marries whom, then we need only require that the number m of women be at least as large as the number n of men, in order to marry off all the men. But we would expect that each man would insist on some compatibility with a spouse and would thereby eliminate some of the women as potential spouses. Thus, each man would arrive at a certain set of acceptable spouses from the available women. (A woman could also eliminate herself from the set of potential spouses of any man whom she finds unacceptable.) Let $(A_1, A_2, \cdots, A_n)$ be the family of subsets of the women, where A_i denotes the set of spouses acceptable to the ith man $(i = 1, \cdots, n)$. Then a *complete marriage* of the men corresponds to an SDR $(w_1, w_2, \cdots, w_n)$ of $(A_1, A_2, \cdots, A_n)$. The correspondence is that the ith man marries the woman w_i, $(i = 1, 2, \cdots, n)$. Since w_i is in A_i, w_i is an acceptable spouse for the ith man.



Systems of distinct representatives IV

Theorem

The family $\mathcal{A} = \{A_1, A_2, \ldots, A_n\}$ of sets have an SDR if and only if the Marriage Condition (MC) holds.

Theorem

Let $\mathcal{A} = \{A_1, A_2, \ldots, A_n\}$ be a family of subsets of a set Y. Then the largest number ρ of sets of $\mathcal{A}$ which can be chosen so that they have an SDR equals the smallest value taken on by the expression

$$|A_{i_1} \cup A_{i_2} \cup \cdots \cup A_{i_k}| + n - k$$

over all choices of $k = 1, 2, \ldots, n$ and all choices of k distinct indices $i_1, i_2, \ldots, i_k$ from $\{1, 2, \ldots, n\}$.



Systems of distinct representatives V

- The number ρ defined in the theorem is the matching number $\rho(G)$ of the bipartite graph G that we have associated with the family $\mathcal{A}$.

Example

A family $\mathcal{A} = (A_1, A_2, A_3, A_4, A_5, A_6)$ of subsets of $\{a, b, c, d, e, f\}$ is defined by

$$A_1 = \{a, b, c\}, A_2 = \{b, c\}, A_3 = \{b, c\},$$
$$A_4 = \{b, c\}, A_5 = \{c\}, A_6 = \{a, b, c, d\}.$$

We have

$$|A_2 \cup A_3 \cup A_4 \cup A_5| = |\{b, c\}| = 2,$$

hence,

$$|A_2 \cup A_3 \cup A_4 \cup A_5| + 6 - 4 = 4.$$



Systems of distinct representatives VI

Thus, we see that at most four of the sets $\mathcal{A}$ can be chosen so that they have an SDR. Since (A_1, A_2, A_5, A_6) has (a, b, c, d) as an SDR, it follows that 4 is the largest number of sets with an SDR.



Stable marriages I

- There are n women and n men in a community. Each woman ranks each man in accordance with her preference for that man as a spouse. No ties are allowed, so that if a woman is indifferent between two men, we nonetheless require that she express some preference. The preferences are to be purely ordinal, and thus each woman ranks the men in the order $1, 2, \ldots, n$. Similarly, each man ranks the women in the order $1, 2, \ldots, n$. There are $n!$ ways in which the women and men can be paired so that a *complete marriage* takes place. We say that a complete marriage is *unstable*, provided that there exist two women A and B and two men a and b such that
  1. A and a get married;
  2. B and b get married;
  3. A prefers (i.e., ranks higher) b to a;
  4. b prefers A to B.
- A complete marriage is called *stable*, provided it is not unstable.



Stable marriages II

- Let $G = \{X, \Delta, Y\}$ be a bipartite graph in which

$$X = \{w_1, w_2, \ldots, w_n\}$$

is the set of n women and

$$Y = \{m_1, m_2, \ldots, m_n\}$$

is the set of n men. We join each woman-vertex (left is now woman) to each man-vertex (right is now man). The resulting bipartite graph is *complete* in the sense that it contains all possible edges between its two sets of vertices. Corresponding to each edge $\{w_i, m_j\}$, there is a pair p, q of numbers where p denotes the position of m_j in w_i's ranking of the men, and q denotes the position of w_i in m_j's ranking of the women. A complete marriage of the women and men corresponds to a perfect matching (of n edges) in this bipartite graph G.



Stable marriages III

- It is more convenient, for notational purposes, to use the model afforded by the *preferential ranking matrix*. This matrix is an n-by-n array of n rows, one for each of the n women $w_1, w_2, \ldots, w_n$, and n columns, one for each of the n men $m_1, m_2, \ldots, m_n$. In the position at the intersection of row i and column j, we place the pair p, q of numbers representing, respectively, the ranking of m_j by w_i and the ranking of w_i by m_j. A complete marriage corresponds to a set of n positions of the matrix that includes exactly one position from each row and one position from each column.



Stable marriages IV

Example

Let $n = 2$, and let the preferential ranking matrix be

$$\begin{array}{c} \\ w_1 \\ w_2 \end{array} \begin{array}{c} \begin{array}{cc} m_1 & m_2 \end{array} \\ \begin{bmatrix} 1,2 & 2,2 \\ 2,1 & 1,1 \end{bmatrix} \end{array}.$$

There are two possible complete marriages,

(1) $w_1 \leftrightarrow m_1, w_2 \leftrightarrow m_2$,

(2) $w_1 \leftrightarrow m_2, w_2 \leftrightarrow m_1$.

The first is readily seen to be stable. The second is unstable.



Stable marriages V

Example

Let $n = 3$, and let the preferential ranking matrix be

$$\begin{bmatrix} 1,3 & 2,2 & 3,1 \\ 3,1 & 1,3 & 2,2 \\ 2,2 & 3,1 & 1,3 \end{bmatrix}. \tag{9.8}$$

There are $3! = 6$ possible complete marriages. One is

$$w_1 \leftrightarrow m_1, w_2 \leftrightarrow m_2, w_3 \leftrightarrow m_3.$$

Since each woman gets her first choice, the complete marriage is stable, even though each man gets his last choice! Another stable complete marriage is obtained by giving each man his first choice.



Stable marriages VI

Theorem

For each preferential ranking matrix, there is a stable complete marriage.

Deferred Acceptance Algorithm

Begin with every woman marked as rejected. While there exists a rejected woman, do the following,

1. Each woman marked as rejected chooses the man whom she ranks highest among all those men who have not yet rejected her.
2. Each man picks out the woman he ranks highest among all those women who have chosen him and whom he has not yet rejected, defers decision on her, and now rejects the others.



Stable marriages VII

Example

We apply the deferred acceptance algorithm to the preferential ranking matrix

	a	b	c	d
A	$1,2$	$2,1$	$3,2$	$4,1$
B	$2,4$	$1,2$	$3,1$	$4,2$
C	$2,1$	$3,3$	$4,3$	$1,4$
D	$1,3$	$4,4$	$3,4$	$2,3$

Hence

$$A \leftrightarrow b, B \leftrightarrow c, C \leftrightarrow a, D \leftrightarrow d$$

is a stable complete marriage.



Stable marriages VII

Example

We apply the deferred acceptance algorithm to the preferential ranking matrix

	a	b	c	d
A	$1,2$	$2,1$	$3,2$	$4,1$
B	$2,4$	$1,2$	$3,1$	$4,2$
C	$2,1$	$3,3$	$4,3$	$1,4$
D	$1,3$	$4,4$	$3,4$	$2,3$

Hence

$$A \leftrightarrow b, B \leftrightarrow c, C \leftrightarrow a, D \leftrightarrow d$$

is a stable complete marriage.



Stable marriages VII

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We apply the deferred acceptance algorithm to the preferential ranking matrix

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D	$1,3$	$4,4$	$3,4$	$2,3$

Hence

$$A \leftrightarrow b, B \leftrightarrow c, C \leftrightarrow a, D \leftrightarrow d$$

is a stable complete marriage.



Stable marriages VII

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D	$1,3$	$4,4$	$3,4$	$2,3$

Hence

$$A \leftrightarrow b, B \leftrightarrow c, C \leftrightarrow a, D \leftrightarrow d$$

is a stable complete marriage.



Stable marriages VII

Example

We apply the deferred acceptance algorithm to the preferential ranking matrix

	a	b	c	d
A	$1,2$	$2,1$	$3,2$	$4,1$
B	$2,4$	$1,2$	$3,1$	$4,2$
C	$2,1$	$3,3$	$4,3$	$1,4$
D	$1,3$	$4,4$	$3,4$	$2,3$

Hence

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is a stable complete marriage.



Stable marriages VII

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We apply the deferred acceptance algorithm to the preferential ranking matrix

	a	b	c	d
A	$1,2$	$2,1$	$3,2$	$4,1$
B	$2,4$	$1,2$	$3,1$	$4,2$
C	$2,1$	$3,3$	$4,3$	$1,4$
D	$1,3$	$4,4$	$3,4$	$2,3$

Hence

$$A \leftrightarrow b, B \leftrightarrow c, C \leftrightarrow a, D \leftrightarrow d$$

is a stable complete marriage.



Stable marriages VII

Example

We apply the deferred acceptance algorithm to the preferential ranking matrix

	a	b	c	d
A	$1,2$	$2,1$	$3,2$	$4,1$
B	$2,4$	$1,2$	$3,1$	$4,2$
C	$2,1$	$3,3$	$4,3$	$1,4$
D	$1,3$	$4,4$	$3,4$	$2,3$

Hence

$$A \leftrightarrow b, B \leftrightarrow c, C \leftrightarrow a, D \leftrightarrow d$$

is a stable complete marriage.



Stable marriages VIII

Example

We apply the deferred acceptance algorithm to the preferential ranking matrix, where the men choose the women.

	a	b	c	d
A	$1,2$	$2,1$	$3,2$	$4,1$
B	$2,4$	$1,2$	$3,1$	$4,2$
C	$2,1$	$3,3$	$4,3$	$1,4$
D	$1,3$	$4,4$	$3,4$	$2,3$

Hence

$$a \leftrightarrow C, b \leftrightarrow A, c \leftrightarrow B, d \leftrightarrow D$$

is a stable complete marriage.



Stable marriages VIII

Example

We apply the deferred acceptance algorithm to the preferential ranking matrix, where the men choose the women.

	a	b	c	d
A	$1,2$	$2,1$	$3,2$	$4,1$
B	$2,4$	$1,2$	$3,1$	$4,2$
C	$2,1$	$3,3$	$4,3$	$1,4$
D	$1,3$	$4,4$	$3,4$	$2,3$

Hence

$$a \leftrightarrow C, b \leftrightarrow A, c \leftrightarrow B, d \leftrightarrow D$$

is a stable complete marriage.



Stable marriages VIII

Example

We apply the deferred acceptance algorithm to the preferential ranking matrix, where the men choose the women.

	a	b	c	d
A	$1,2$	$2,1$	$3,2$	$4,1$
B	$2,4$	$1,2$	$3,1$	$4,2$
C	$2,1$	$3,3$	$4,3$	$1,4$
D	$1,3$	$4,4$	$3,4$	$2,3$

Hence

$$a \leftrightarrow C, b \leftrightarrow A, c \leftrightarrow B, d \leftrightarrow D$$

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Stable marriages VIII

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Hence

$$a \leftrightarrow C, b \leftrightarrow A, c \leftrightarrow B, d \leftrightarrow D$$

is a stable complete marriage.



Stable marriages IX

A stable complete marriage is called *optimal for a woman*, provided that a woman gets as a spouse a man whom she ranks at least as high as the spouse she obtains in every other stable complete marriage. A stable marriage is called *women-optimal* provided that it is optimal for each woman.

Theorem

The stable complete marriage obtained from the deferred acceptance algorithm, with the women choosing the men, is women-optimal. If the men choose women in the deferred acceptance algorithm, the resulting complete marriage is men-optimal.

Corollary

In the women-optimal stable complete marriage, each man is paired with the woman he ranks lowest among all the partners that are possible for him in a stable complete marriage.

